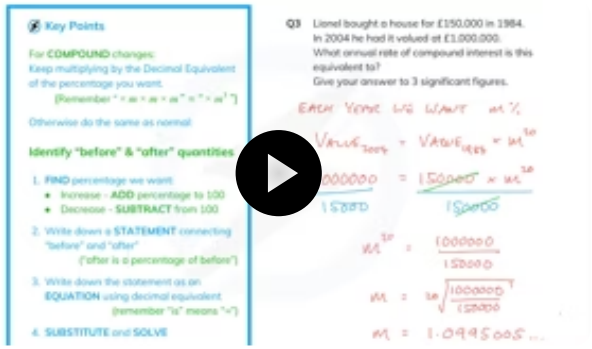
Compound Interest

Exam code: 0607

Download PDF

Written by: Jamie Wood
Reviewed by: Dan Finlay
Updated on 19 January 2026



Did this video help you? Yes No

Compound Interest

What is compound interest?

- **Compound interest** is where interest is calculated on the **running total**, not just the starting amount
  - This is different from **simple interest** where interest is only based on the starting amount
- e.g. **$ 100** earns **10% interest** each year, for 3 years
  - At the end of year 1, **10% of $ 100 is earned**
    - The total balance will now be 100+10 = **$ 110**
  - At the end of year 2, **10% of $ 110 is earned**
    - The balance will now be 110+11 = **$ 121**
  - At the end of year 3, **10% of $ 121 is earned**
    - The balance will now be 121+12.1 = **$ 133.10**

How do I calculate compound interest?

- Compound interest **increases an amount by a percentage**, and then **increases the new amount by the same percentage**
  - This process repeats each time period (yearly or monthly etc)
- We can use a **multiplier** to carry out the percentage increase multiple times

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

would find (300×1.05)×1.05
        - This could be rewritten as **300×1.05^2**
    - To increase **$ 300** by **5%**, each year for **3 years**, we would find ((300×1.05)×1.05)×1.05
        - This could be rewritten as **300×1.05^3**
- This can be extended to any number of periods that the interest is applied for
    - If **$ 2000** is subject to **4%** compound interest each year for **12 years**
    - Find **2000×1.04^{12}**, which is $ 3202.06
- Note that this method calculates the **total balance** at the end of the period, **not the interest earned**

How do I calculate depreciation?

- A similar method can be used if something **decreases in value** by a percentage every year (e.g. a car)
- This is known as **depreciation**
- Change the **multiplier** to one which represents a percentage decrease
    - e.g. a decrease of 15% would be a multiplier of 0.85
- If a car worth $ 16 000 depreciates by 15% each year for 6 years
    - Its value will be $16\,000 \times 0.85^6$, which is $ 6034.39

Compound interest formula

- An **alternative method** is to use the **following formula** to calculate the final balance
    - Final balance = $P\left(1+\frac{r}{100}\right)^n$ where
        - P is the original amount,
        - r is the % increase,
        - and n is the number of years
    - Note that $1+\frac{r}{100}$ is the same value as the multiplier
        - e.g. 1.15 for 15% interest
- This formula is **not given** in the exam

Examiner Tips and Tricks

- Double check if the question uses simple interest or compound interest
- The formula for compound interest is **not** given in the exam

Worked Example

To the nearest dollar, what is her investment worth at the end of the 7 years?

Answer:

We want an increase of 4% per year, this is equivalent to a multiplier of 1.04, or 104% of the original amount

This multiplier is applied 7 times;
$\times 1.04 \times 1.04 \times 1.04 \times 1.04 \times 1.04 \times 1.04 \times 1.04 = 1.04^7$

Therefore the final value after 7 years will be

$$1200 \times 1.04^7 = \$\ 1579.118135$$

Round to the nearest dollar

$$\$\ 1579$$

Alternate method

Using the formula for the final amount $P\left(1 + \frac{r}{100}\right)^n$

Substitute P is 1200, $r = 4$ and $n = 7$ into the formula

$$1200\left(1 + \frac{4}{100}\right)^7$$

$$\$\ 1579$$

Teacher only

Test your students on this in just a few clicks

- Create exam-aligned tests
- Tailor format & difficulty
- Quizzes or homework in minutes

Try Test Builder →

Test yourself

Was this revision note helpful? Yes No

Resources

Learning Hub

Scholarship

Join

Past Papers

Solution Banks

Sitemap

Dashboard

Overview

Brand ambassador

Profile

Subscription

Settings

Log out

Company

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Cookie Policy

Help and Support

Subjects

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.